Chapter - 7
Normal distribution

A random variable X is said to have a normal distribution with mean $\mu (-\infty < \mu < \infty)$ and variance $\sigma^2 > 0$ if it has the density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(\frac{1}{2})[(x-\mu)/\sigma]^2} \quad -\infty < x < \infty$$

Properties -

1) $\int_{-\infty}^{\infty} f(x)\,dx = 1$
2) $f(x) \geq 0$

} required of all density function

3) $\lim_{x \to \pm\infty} f(x) = 0$

4) $f(\mu + x) = f(\mu - x)$

5) the maximum value of f occurs at $x = \mu$

6) the points of inflection of f are at $x = \mu \pm \sigma$

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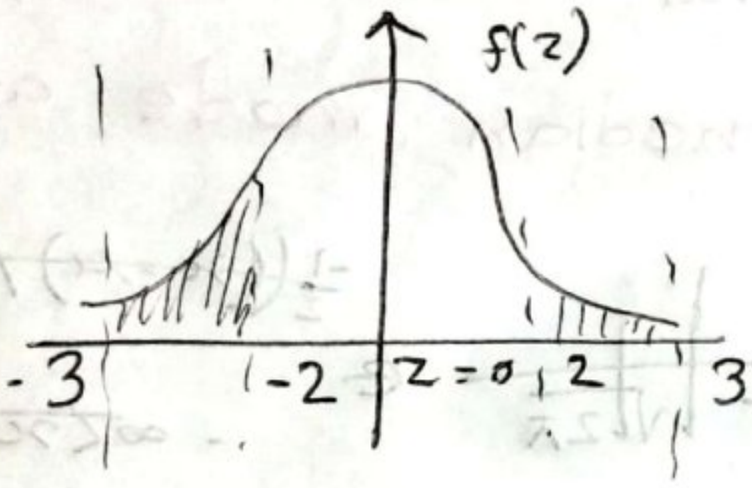
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The standard Normal Distribution

Z is standard normal distribution with mean zero and variance 1.

Let $Z = \frac{x-\mu}{\sigma}$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}} \qquad -\infty < z < \infty$$



$$p(z \leq -2) = \phi(-2)$$

If the population ~~of~~ distribution of a variable is normal,

1. Roughly 68% of the values are within 1 SD of mean
2) 95% - - - - 2 SDs
3) 99.7% - - - - 3 SDs

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~~$P(a \leq x \leq b) = P($~~

$P(X < 60)$

$= P\left(\frac{X-\mu}{\sigma} \leq \frac{60-\mu}{\sigma}\right)$

* What is Pdf of normal distribution and standard normal distribution? Show that mean, median, mode are equal?

~~$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}((x-\mu)/\sigma)^2} \quad -\infty < x < \infty$~~

A random variable ~~of~~ X is said to have a normal distribution with mean μ $(-\infty < \mu < \infty)$ and variance $\sigma^2 > 0$ if the Pdf is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}((x-\mu)/\sigma)^2} \quad \text{for } -\infty < x < \infty$$

A continuous random variable Z is said to have a standard normal distribution with mean 0 and variance 1 if the pdf is

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$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad \text{for} \quad -\infty < z < \infty$$

Mean

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$$

$$= \int_{-\infty}^{\infty} x \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2}\left((x-\mu)/\sigma\right)^2}$$

$$= \int_{-\infty}^{\infty} \frac{(\mu+\sigma z)}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz \qquad \text{let} \quad z = \frac{x-\mu}{\sigma}$$

$$dz = \frac{1}{\sigma} dx$$

$$x = \mu + \sigma z$$

$$= \mu \int_{-\infty}^{\infty} \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi}} dz + \sigma \int_{-\infty}^{\infty} \frac{z}{\sqrt{2\pi}} e^{-z^2/2} dz$$

Since the integrand of the first ~~normal~~ integral is that of a normal density with $\mu = 0$ and $\sigma^2 = 1$, the ~~integ~~ value of first integral is 1.

The second integral has value zero, that is

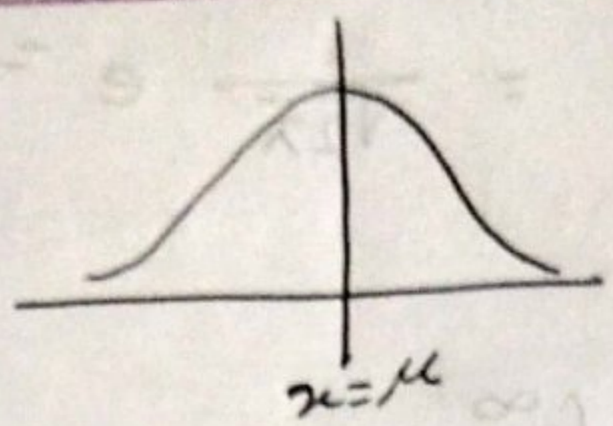
$$\int_{-\infty}^{\infty} \frac{z e^{\frac{-z^2}{2}}}{\sqrt{2\pi}} dz = \left[\frac{-e^{\frac{-z^2}{2}}}{\sqrt{2\pi}}\right]_0^{\infty} = 0$$

$$\therefore \quad E(X) = \mu[1] + 0 = \mu$$

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Median:



Let M be the median

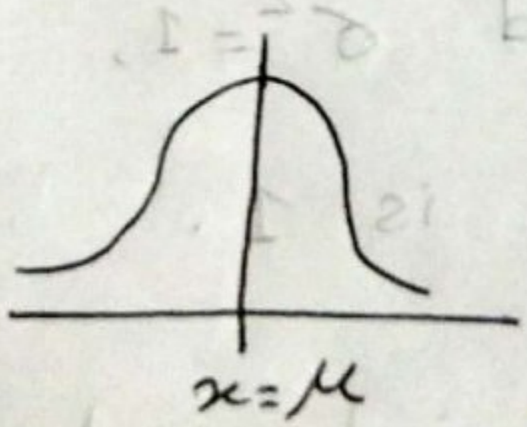
$$P(X \leq M) = P(X \geq M)$$

$$\int_{-\infty}^{M} f(x)\,dx = \int_{M}^{\infty} f(x)\,dx = \frac{1}{2}$$

$$M = \mu$$

Mode

Mode is the value of x for which the probability is maximum.



$\therefore$ mode $= \mu$

So the mean, = median = mode $= \mu$.

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The article of "Monte Carlo Simulation - Tool ----" suggest that yield strength (ksi) for A36 steel grade is normally distributed with $\mu = 43$ and $\sigma = 4.50$

what is the probability that yield strength is

i) at most 40

ii) Greater than 60

iii) Between 35 to 55

i) $P(X \leq 40)$

$= P\left(\frac{X-\mu}{\sigma} \leq \frac{40-43}{4.5}\right)$

$= P(Z \leq -0.66)$

$= \phi(-0.66)$

$= 0.25463$

ii) $P(X > 60)$

$= 1 - P(X \leq 60)$

$= 1 - P\left(\frac{X-\mu}{\sigma} \leq \frac{60-43}{4.5}\right)$

$= 1 - 0.99992$

$= 0.00008$

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iii) $P(35 < x < 55)$

$P[a < x < b]$

$= F(b) - F(a)$

$= P\left(\frac{35-\mu}{\sigma} < \frac{x-\mu}{\sigma} < \frac{55-\mu}{\sigma}\right)$

$= P(-1.78 \leq z < 2.67)$

$= \phi(2.67) - \phi(-1.78)$

$= 0.99621 - 0.03754$

$= 0.95867$

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